CALPHAD: Computer Coupling of Phase Diagrams and Thermochemistry 35 (2011) 551–555

Contents lists available at SciVerse ScienceDirect

CALPHAD: Computer Coupling of Phase Diagrams and Thermochemistry

journal homepage: www.elsevier.com/locate/calphad

Thermodynamic assessment of the Ce–Pt system

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ARTICLE INFO

Article history:
Received 8 April 2011
Received in revised form
20 September 2011
Accepted 24 September 2011
Available online 25 October 2011

Keywords:
Phase diagrams
Rare earth alloys and compounds
Thermodynamic modeling

ABSTRACT

The thermodynamic assessment of the Ce–Pt binary system has been carried out by using the CALPHAD (Calculation of Phase Diagrams) method based on the available literature data including the phase equilibria and thermodynamic properties. The Gibbs free energies of the liquid, bcc_A2 and fcc_A1 phases were described by the subregular solution model with the Redlich–Kister formula, and those of the intermetallic compounds (Ce_7Pt_3, Ce_3Pt_2, Ce_5Pt_4, CePt, Ce_3Pt_4, $CePt_x$ and $CePt_5$) in the Ce–Pt binary system were described by the sublattice model, and two intermetallic compounds (Ce_5Pt_3, $CePt_2$) in the Ce–Pt binary system were described by the two-sublattice model. A consistent set of thermodynamic parameters leading to a reasonable agreement between the calculated results and literature data was obtained.

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1. Introduction

The rare earth elements exhibiting a good combination of magnetic, optical, electrical and thermal properties have been paid increasing attention and widely used to prepare new high-performance functional materials, such as permanent magnet, magnetic refrigerant and hydrogen storage material [1–3]. In order to develop high-performance materials with rare earth elements, alloying of rare earth elements and other metals is of both industrial demand and technical interest. Thus, accurate information on the phase diagrams of rare earth alloy systems, which can not only help to effectively select the alloying method but also help to well understand the relationship between the composition and the microstructure, is especially required. Obtaining information on the phase diagrams exclusively from experiments is burdensome and expensive. The CALPHAD technique, which has been recognized to be an important tool to significantly reduce time and cost during the development of materials, can provide a clear guidance for the materials design [4–6]. In order to develop the thermodynamic database of the rare earth alloys for reliable predictions of liquidus, phase fraction, equilibrium and non-equilibrium solidification behavior, etc. in multi-component systems, our research group has made some thermodynamic assessments in the rare earth alloy systems [7–12]. In the present work, as a part of the above-mentioned thermodynamic database, the thermodynamic assessment of the Ce–Pt binary system was carried out by means of the CALPHAD method.

2. Literature data

Regarding to the information on phase equilibria and phase transformations, Vronova et al. [13] reported that the temperature of the eutectic reaction ($L \leftrightarrow CePt_5 + (Pt)$) and its eutectic point are 1575 °C and 92 at.% Pt, respectively. Janghorban et al. [14] investigated the Ce–Pt phase diagram over the whole composition range by means of differential thermal analysis (DTA), X-ray diffraction (XRD), optical microscopy, scanning electron microscopy (SEM), energy dispersive X-ray analysis (EDX) and micro-hardness measurements, and confirmed the existence of nine intermediate phases (Ce_7Pt_3, Ce_3Pt_2, CePt, Ce_3Pt_4, $CePt_2$, $CePt_5$, Ce_5Pt_3, Ce_5Pt_4 and $CePt_x$ ($x \approx 4.5$)) in the Ce–Pt system. The phase diagram of the Ce–Pt system is shown in Fig. 1 from [14].

Regarding to the thermodynamic properties, the enthalpies of formation of the $CePt_2$ and CePt phases at 298 K with respect to the fcc_A1 (γCe) and fcc-A1 (Pt) phases were measured by Sellhaoui and Kleppa [15] using reaction and drop calorimetry, and the enthalpy of formation of the CePt phase at 833 K with respect to the fcc_A1 (γCe) and fcc-A1 (Pt) phases was reported by Palenzona and Cirafici [16] by using the dynamic differential calorimeter data method. Kemmler [17] investigated the enthalpy and entropy of formation of the $CePt_5$ phase from 1073 to 1373 K on the basis of Electromotive force (EMF) measurement. In addition, the enthalpy of mixing of liquid phase in the Ce–Pt binary system calculated by Takeuchi and Inoue [18] is −81 kJ/mol at 50 at.% Pt, however, no temperature dependence was given in the work of Takeuchi and Inoue [18].

All the above-mentioned literature data including phase equilibria and thermodynamic properties will be used in the following thermodynamic assessment of the Ce–Pt system.

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doi:10.1016/j.calphad.2011.09.006

Table 1
The stable solid phases and the used models in the Ce–Pt system.

Phase	Composition (at.% Pt)	Prototype	Strukturbericht symbol	Modeling phase	Used models
(δCe)	0	W	A2	$(Ce, Pt)_1Va_1$	SSM
(γCe)	0	Cu	A1	$(Ce, Pt)_1Va_3$	SSM
Ce_7Pt_3	30	Th_7Fe_3	$D10_2$	$(Ce)_{0.7}(Pt)_{0.3}$	SM
Ce_5Pt_3	37.5–38. 5	Pu_5Rh_3	–	$(Ce, Pt)_{0.615}(Ce, Pt)_{0.385}$	TSM
Ce_3Pt_2	40	Er_3Ni_2	–	$(Ce)_{0.6}(Pt)_{0.4}$	SM
Ce_5Pt_4	44.4	Ge_4Sm_5	–	$(Ce)_{0.5555}(Pt)_{0.4444}$	SM
CePt	50	CrB	B_f	$(Ce)_{0.5}(Pt)_{0.5}$	SM
Ce_3Pt_4	57.14	Pu_3Pd_4	–	$(Ce)_{0.4286}(Pt)_{0.5714}$	SM
$CePt_2$	66.67–75	Cu_2Mg	C15	$(Ce, Pt)_{0.325}(Ce, Pt)_{0.675}$	TSM
$CePt_x$	∼82	–	–	$(Ce)_{0.18}(Pt)_{0.82}$	SM
$CePt_5$	83.33	$CaCu_5$	$D2_d$	$(Ce)_{0.1666}(Pt)_{0.8333}$	SM
(Pt)	100	Cu	A1	$(Ce, Pt)_1Va_3$	SSM

SSM: subregular solution model; SM: sublattice model; TSM: two-sublattice model.

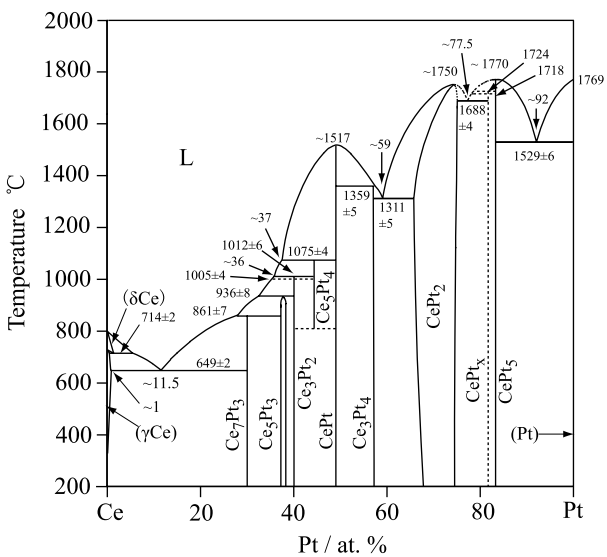

Fig. 1. The phase diagram of the Ce–Pt system reviewed by Janghorban et al. [14].

3. Thermodynamic models

The related information of stable solid phases and the models used in the Ce–Pt system are listed in Table 1.

3.1. Solution phases

In the Ce–Pt binary system, Gibbs free energies of the liquid, Bcc_A2 and Fcc_A1 are described by the subregular solution model with the Redlich–Kister formula [19]. The molar Gibbs free energies of the solution phase in the Ce–Pt binary system are given as follows:

$$G_m^{\phi} = \sum_{i=\mathrm{Ce,Pt}} {}^0G_i^{\phi} x_i + RT \sum_{i=\mathrm{Ce,Pt}} x_i \ln x_i + x_{\mathrm{Ce}} x_{\mathrm{Pt}} \sum_{j=0}^{n} {}^jL_{\mathrm{Ce,Pt}}^{\phi} (x_{\mathrm{Ce}} - x_{\mathrm{Pt}})^j \quad (1)$$

where ${}^0G_i^{\phi}$ is the Gibbs free energy of the pure component i in the respective reference state with the ϕ phase, which is taken from the SGTE pure element database [20], x_i represents the mole fraction of the component i; R is the gas constant and T is the absolute temperature; ${}^jL_{\mathrm{Ce,Pt}}^{\phi}$ is the binary interaction parameter and is expressed as:

$${}^jL_{\mathrm{Ce,Pt}}^{\phi} = a_j + b_j T \quad (2)$$

where the values of a_j and b_j were evaluated in the present work based on the literature data.

Table 2
The optimized thermodynamic parameters of the Ce–Pt system.

Parameters in each phase (J/mol)
Liquid phase, format $(Ce, Pt)_1$
${}^0L_{\mathrm{Ce,Pt}}^{Liq} = -370000 + 20T$
${}^1L_{\mathrm{Ce,Pt}}^{Liq} = 135500 - 20.1T$
${}^2L_{\mathrm{Ce,Pt}}^{Liq} = 10000 + 30.2T$
${}^3L_{\mathrm{Ce,Pt}}^{Liq} = -23600$
Bcc_A2 phase, format $(Ce, Pt)_1$
${}^0L_{\mathrm{Ce,Pt}}^{\mathrm{Bcc_A2}} = -232000 + 20.4T$
Fcc_A1 phase, format $(Ce, Pt)_1$
${}^0L_{\mathrm{Ce,Pt}}^{\mathrm{Fcc_A1}} = -205000 + 10T$
Ce_7Pt_3 phase, format $(Ce)_{0.7}(Pt)_{0.3}$
$G_{\mathrm{Ce:Pt}}^{\mathrm{Ce_7Pt_3}} = -74920 + 8.04T + 0.7{}^0G_{\mathrm{Ce}}^{\mathrm{Fcc}} + 0.3{}^0G_{\mathrm{Pt}}^{\mathrm{Fcc}}$
Ce_5Pt_3 phase, format $(Ce, Pt)_{0.615}(Ce, Pt)_{0.385}$
$G_{\mathrm{Ce:Pt}}^{\mathrm{Ce_5Pt_3}} = -88000 + 5.997T + 0.615{}^0G_{\mathrm{Ce}}^{\mathrm{Fcc}} + 0.385\ {}^0G_{\mathrm{Pt}}^{\mathrm{Fcc}}$
$G_{\mathrm{Ce:Ce}}^{\mathrm{Ce_5Pt_3}} = 13600 + 17T + {}^0G_{\mathrm{Ce}}^{\mathrm{Fcc}}$
$G_{\mathrm{Pt:Pt}}^{\mathrm{Ce_5Pt_3}} = 3000 + {}^0G_{\mathrm{Pt}}^{\mathrm{Fcc}}$
${}^0L_{\mathrm{Ce,Pt:Pt}}^{\mathrm{Ce_5Pt_3}} = -120000$
${}^0L_{\mathrm{Ce:Ce,\ Pt}}^{\mathrm{Ce_5Pt_3}} = -43690 + 9.5T$
Ce_3Pt_2 phase, format $(Ce)_{0.6}(Pt)_{0.4}$
$G_{\mathrm{Ce:Pt}}^{\mathrm{Ce_3Pt_2}} = -90000 + 5.538T + 0.6{}^0G_{\mathrm{Ce}}^{\mathrm{Fcc}} + 0.4{}^0G_{\mathrm{Pt}}^{\mathrm{Fcc}}$
Ce_5Pt_4 phase, format $(Ce)_{0.555}(Pt)_{0.444}$
$G_{\mathrm{Ce:Pt}}^{\mathrm{Ce_5Pt_4}} = -94460 + 3.7959T + 0.555{}^0G_{\mathrm{Ce}}^{\mathrm{Fcc}} + 0.444{}^0G_{\mathrm{Pt}}^{\mathrm{Fcc}}$
CePt phase, format $(Ce)_{0.5}(Pt)_{0.5}$
$G_{\mathrm{Ce:Pt}}^{\mathrm{CePt}} = -102800 + 4.165T + 0.5{}^0G_{\mathrm{Ce}}^{\mathrm{Fcc}} + 0.5{}^0G_{\mathrm{Pt}}^{\mathrm{Fcc}}$
Ce_3Pt_4 phase, format $(Ce)_{0.4286}(Pt)_{0.5714}$
$G_{\mathrm{Ce:Pt}}^{\mathrm{Ce_3Pt_4}} = -99343 + 2.92T + 0.4286 + {}^0G_{\mathrm{Ce}}^{\mathrm{Fcc}} + 0.5714{}^0G_{\mathrm{Pt}}^{\mathrm{Fcc}}$
$CePt_2$ phase, format $(Ce,\ Pt)_{0.325}(Ce,\ Pt)_{0.675}$
$G_{\mathrm{Ce:Pt}}^{\mathrm{CePt_2}} = -90600 + 1.945T + 0.325{}^0G_{\mathrm{Ce}}^{\mathrm{Fcc}} + 0.675{}^0G_{\mathrm{Pt}}^{\mathrm{Fcc}}$
$G_{\mathrm{Ce:Ce}}^{\mathrm{CePt_2}} = 68000 + 20T + {}^0G_{\mathrm{Ce}}^{\mathrm{Fcc}}$
$G_{\mathrm{Pt:Pt}}^{\mathrm{CePt_2}} = 45000 + {}^0G_{\mathrm{Pt}}^{\mathrm{Fcc}}$
${}^0L_{\mathrm{Ce,Pt:Pt}}^{\mathrm{CePt_2}} = -237000 + 20T$
${}^0L_{\mathrm{Ce:Ce,Pt}}^{\mathrm{CePt_2}} = -140210 + 21T$
$CePt_x$ phase, format $(Ce)_{0.18}(Pt)_{0.82}$
$G_{\mathrm{Ce:Pt}}^{\mathrm{CePt_x}} = -74165 + 6.301T + 0.18{}^0G_{\mathrm{Ce}}^{\mathrm{Fcc}} + 0.82{}^0G_{\mathrm{Pt}}^{\mathrm{Fcc}}$
$CePt_5$ phase, format $(Ce)_{0.1666}(Pt)_{0.8333}$
$G_{\mathrm{Ce:Pt}}^{\mathrm{CePt_5}} = -71100 + 6.41T + 0.1666{}^0G_{\mathrm{Ce}}^{\mathrm{Fcc}} + 0.8333{}^0G_{\mathrm{Pt}}^{\mathrm{Fcc}}$

3.2. Stoichiometric intermetallic compounds

The intermetallic compounds (Ce_7Pt_3, Ce_3Pt_2, Ce_5Pt_4, CePt, Ce_3Pt_4, $CePt_x$, $CePt_5$) in the Ce–Pt binary system are treated as the

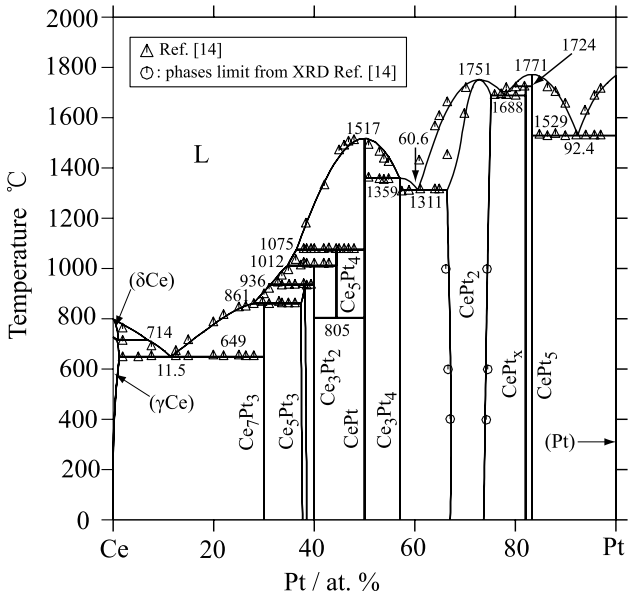

Fig. 2. The calculated phase diagram in the Ce–Pt system compared with the experimental data [14].

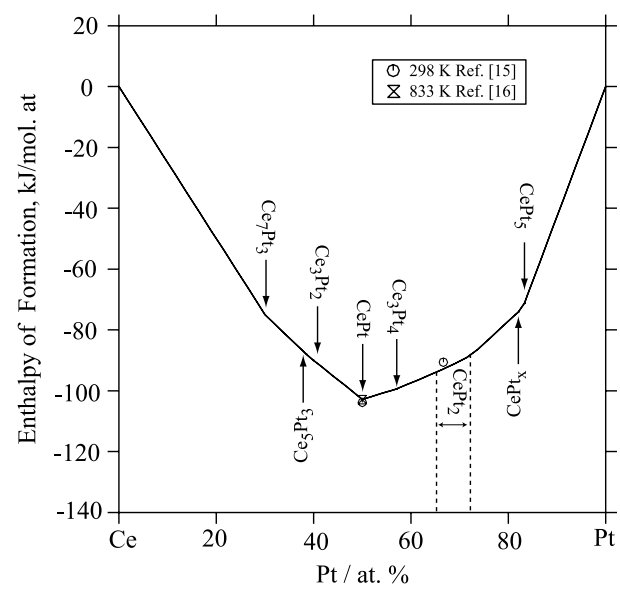

Fig. 3. The calculated enthalpies of formation of intermetallic phases at 833 K compared with the experimental data [15,16].

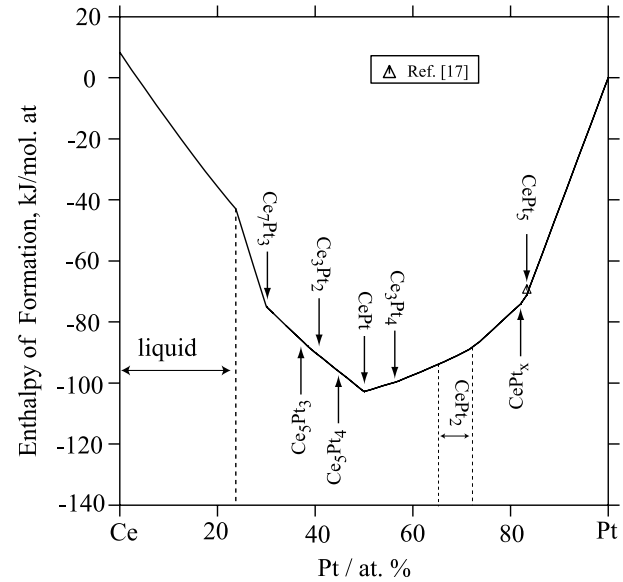

Fig. 4. The calculated enthalpy of formation of intermetallic phases at 1100 K compared with the experimental data [17].

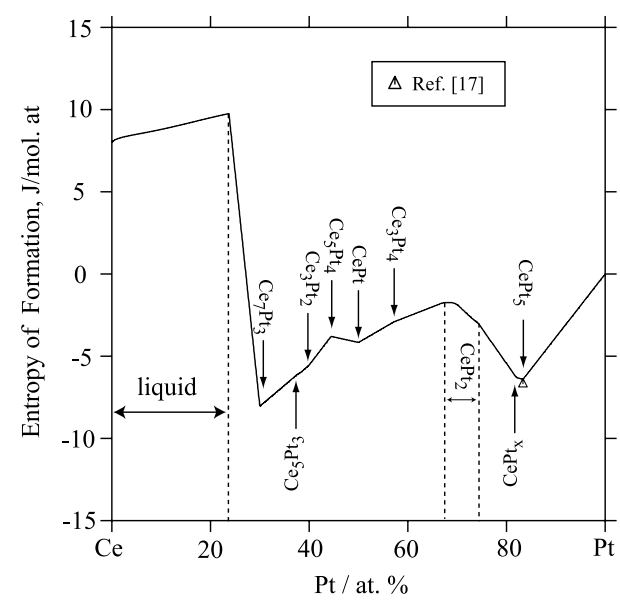

Fig. 5. The calculated entropy of formation of intermetallic phases at 1100 K compared with the experimental data [17].

stoichiometric compound described by the sublattice model. The Gibbs free energy per mole of formula unit Ce_pPt_q can be expressed as follows:

$$G^{Ce_pPt_q}_{Ce:Pt} = \Delta^0 G_f^{Ce_pPt_q} + p^0 G^{ref}_{Ce} + q^0 G^{ref}_{Pt} \quad (3)$$

where $\Delta^0 G_f^{Ce_pPt_q}$ indicates the standard Gibbs free energy of formation of the stoichiometric compound from pure elements, which is formulated as:

$$\Delta^0 G_f^{Ce_pPt_q} = a_1 + b_1 T \quad (4)$$

where the parameters of a_1 and b_1 were evaluated in the present work.

3.3. Non-stoichiometric intermetallic phases

The Ce_5Pt_3 and $CePt_2$ phases in the Ce–Pt binary system were treated as a non-stoichiometric intermetallic compound. The Gibbs free energy of the Ce_5Pt_3 and $CePt_2$ phases was described by a two-sublattice model [21]. The Gibbs free energy of formation per mole of formula unit $(Ce, Pt)_p(Ce, Pt)_q$ can be expressed as the following equation with reference to the pure elements in their non-magnetic state:

$$\begin{aligned} G_m^{Ce_pPt_q} = {} & y^I_{Ce} y^{II}_{Ce}\, {}^0G_{Ce:Ce} + y^I_{Ce} y^{II}_{Pt}\, {}^0G_{Ce:Pt} \\ & + y^I_{Pt} y^{II}_{Ce}\, {}^0G_{Pt:Ce} + y^I_{Pt} y^{II}_{Pt}\, {}^0G_{Pt:Pt} \\ & + pRT\left(y^I_{Ce}\ln y^I_{Ce} + y^I_{Pt}\ln y^I_{Pt}\right) \\ & + qRT\left(y^{II}_{Ce}\ln y^{II}_{Ce} + y^{II}_{Pt}\ln y^{II}_{Pt}\right) \\ & + p y^I_{Ce} y^I_{Pt}\left[y^{II}_{Ce}\sum_n {}^nL_{Ce,Pt:Ce}(y^I_{Ce} - y^I_{Pt})^n \right. \\ & \left. + \; y^{II}_{Pt}\sum_n {}^nL_{Ce,Pt:Pt}(y^I_{Ce} - y^I_{Pt})^n \right] \end{aligned}$$

Table 3
The calculated invariant reactions in the Ce–Pt system compared with the literature data.

Reaction type	Reaction	Pt (at.%)			T (°C)	Reference
Congruent	$L \leftrightarrow CePt_5$	0.795	0.795		1770	[14]
		0.795	0.795		1771	This work
Congruent	$L \leftrightarrow CePt_2$	0.75	0.75		1750	[14]
		0.73	0.73		1751	This work
Peritectic reaction	$L + CePt_5 \leftrightarrow CePt_x$	–	0.833	0.82	1724	[14]
		0.795	0.833	0.82	1724.29	This work
Eutectic reaction	$L \leftrightarrow CePt_2 + CePt_x$	0.775	0.752	0.82	1688 ± 4	[14]
		0.779	0.752	0.82	1687.83	This work
Eutectic reaction	$L \leftrightarrow CePt_5 + (Pt)$	0.92	0.833	1	1529	[14]
		0.924	0.833	1	1528.98	This work
Congruent	$L \leftrightarrow CePt$	~0.5	~0.5		~1517	[14]
		0.5	0.5		1517	This work
Eutectic reaction	$L \leftrightarrow CePt + Ce_3Pt_4$	–	~0.5	0.5714	1359 ± 5	[14]
		0.57	0.5	0.5714	1359.18	This work
Eutectic reaction	$L \leftrightarrow Ce_3Pt_4 + CePt_2$	0.59	0.5714	0.667	1311 ± 5	[14]
		0.606	0.5714	0.667	1310.92	This work
Peritectic reaction	$L + CePt \leftrightarrow Ce_5Pt_4$	~0.37	~0.5	0.444	1075 ± 4	[14]
		0.365	~0.5	0.444	1075	This work
Peritectic reaction	$L + Ce_5Pt_4 \leftrightarrow Ce_3Pt_2$	~0.36	0.444	0.4	1012 ± 4	[14]
		0.347	0.444	0.4	1012	This work
Peritectic reaction	$L + Ce_3Pt_2 \leftrightarrow Ce_5Pt_3$	–	0.4	0.382	936 ± 8	[14]
		0.313	0.4	0.382	936	This work
Peritectic reaction	$L + Ce_5Pt_3 \leftrightarrow Ce_7Pt_3$	–	0.375	0.3	861 ± 7	[14]
		0.279	0.375	0.3	861	This work
Eutectoid reaction	$Ce_5Pt_4 \leftrightarrow Ce_3Pt_2 + CePt$	0.444	0.4	0.49	–	[14]
		0.444	0.4	0.5	804	This work
Eutectoid reaction	$(\delta Ce) \leftrightarrow L + (\gamma Ce)$	–	–	<0.2	714 ± 2	[14]
		0.011	0.07	0.008	713.9	This work
Eutectic reaction	$L \leftrightarrow (\gamma Ce) + Ce_7Pt_3$	0.115	0.01	0.3	649 ± 2	[14]
		0.115	0.012	0.3	649	This work

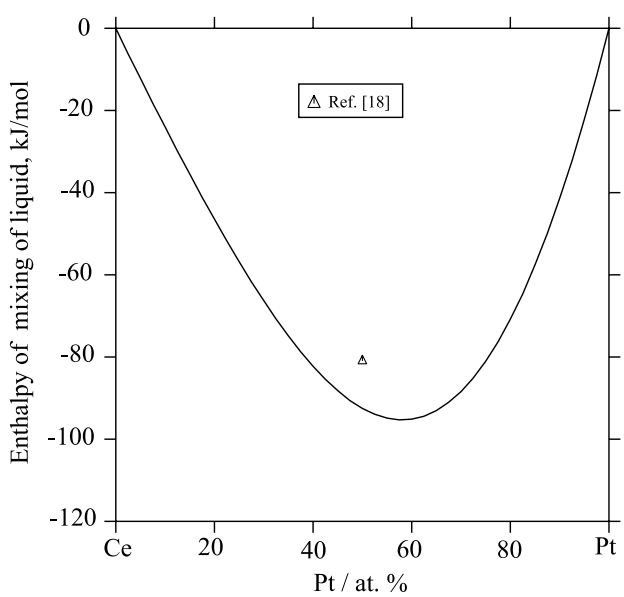

Fig. 6. The calculated enthalpies of mixing of liquid phase at 2073 K with reference states of pure elements of Ce (liquid) and Pt (liquid) compared with the predicted data [18].

$$+ q y^{II}_{Ce} y^{II}_{Pt} \left[y^{I}_{Ce} \sum_n {}^n L_{Ce:Ce,Pt} (y^{II}_{Ce} - y^{II}_{Pt})^n + y^{I}_{Pt} \sum_n {}^n L_{Pt:Ce,Pt} (y^{II}_{Ce} - y^{II}_{Pt})^n \right], \quad (5)$$

where y^{I}_{i} and y^{II}_{j} are the site fractions of component i and j (i, j = Ce, Pt) located on sublattice I and II, respectively, and the parameter ${}^0G_{i:j}$ represents the Gibbs free energy of the compound phase when the two sublattices are occupied by element i or j. $L_{i:Ce,Pt}$ and $L_{Ce,Pt:j}$ are the interaction parameters between Ce and Pt in the second or first sublattice, when the other sublattice is occupied by element i or j. ${}^0G_{i:j}$, $L_{i:Ce,Pt}$ and $L_{Ce,Pt:j}$ were evaluated in the present work.

4. Results and discussion

The thermodynamic parameters of each phase in the Ce–Pt system were optimized using the PARROT [22] module of THERMO-CALC [23]. For the PARROT module in the THERMO-CALC software, the literature data were used as input to the program. Each piece of selected information was given a certain weight by the importance of data, and changed by trial and error during the assessment, until most of the selected data is reproduced within the expected uncertainty limits. In the present work, all the mentioned literature data [14–18] were used in the optimization process. All the parameters were eventually optimized to obtain the best consistency between the calculated results and literature data. All the optimized thermodynamic parameters of each phase in the Ce–Pt system are listed in Table 2.

The calculated phase diagram of the Ce–Pt system compared with the experimental data is shown in Fig. 2. The calculated phase diagram is in a good agreement with the phase diagram investigated by Janghorban et al. [14]. The comparison of the calculated invariant reactions in the Ce–Pt system with the literature data is listed in Table 3, where a reasonable agreement between the calculated and experimental temperatures for those invariant reactions was obtained.

The enthalpies of formation of the intermetallic compounds in the Ce–Pt system were calculated, and presented in Fig. 3. The calculated results are consistently with the literature data in Refs. [15,16]. In addition, the calculated enthalpy and entropy at 1100 K in the Ce–Pt system are in an excellent agreement with the reported data [17], as shown in Figs. 4 and 5, respectively.

Moreover, the calculated enthalpy of mixing of liquid phase at 2073 K with reference states of pure elements of Ce (liquid) and Pt (liquid) in the Ce–Pt system is presented in Fig. 6.

5. Conclusion

Based on the available literature data including the phase equilibria and thermodynamic properties, the thermodynamic assessment of the Ce–Pt binary system has been carried out by means of the CALPHAD technique. A consistent set of optimized thermodynamic parameters has been derived for describing the Gibbs free energy of each phase in the Ce–Pt binary systems, and a good agreement between the calculated results and most of the experimental data has been obtained.

Acknowledgments

This work was supported by the National Natural Science Foundation of China (No. 51031003), the Ministry of Education, PR China (No. 707037), the Ministry of Science and Technology, PR China (No. 2009AA03Z101), and Fujian Province and Xiamen City Departments of Science and Technology (Nos. 2009I0024 and 3502Z20093001).

Appendix. Supplementary data

Supplementary material related to this article can be found online at doi:10.1016/j.calphad.2011.09.006.

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